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The overlooked role of putative non-Hg methylators in predicting methylmercury production in paddy soils

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ABSTRACT

Rice ingestion has been recognized as an important route of dietary exposure to neurotoxic methylmercury (MeHg) that is commonly synthesized in rice paddy soils. Although Hg methylators are known to regulate soil MeHg formation, the effect of non-Hg methylating communities on MeHg production remains unclear. Here, we collected 141 paddy soil samples from main rice-producing areas across China to identify associations between bacterial community composition (including both Hg and putative non-Hg methylators) and MeHg production. Results showed that the MeHg content in the paddy soils varied from 0.11 to 8.36 ng g^{-1} at a national spatial scale, which could be owing to the shifts of soil microbial community composition across different areas. Our structure equation modelling suggested a strong link between bacterial community composition and MeHg content and %MeHg. More importantly, random forest analyses suggested a more significant role of putative non-Hg methylators than Hg methylators in predicting variations of soil MeHg content. The relative abundance of putative non-Hg methylators such as unclassified *Xanthomonadales* and *Chitinophagaceae* were strongly correlated with soil MeHg contents. Further, microbial network analysis revealed strong co-occurrence patterns between putative the non-Hg and Hg methylators. These findings highlight an overlooked role of non-Hg methylating communities in predicting MeHg production in paddy soils.

INTRODUCTION

Methylmercury (MeHg) is a highly neurotoxic compound that is easily accumulated along food chains [1-2]. Rice ingestion has been considered to be an important route of dietary exposure to MeHg [3-4], particularly in Hg-impacted areas where rice is a staple food. MeHg accumulation in rice is due to microbial methylation of inorganic Hg(II) in the paddy soil [5-7], rather than the synthesis of MeHg in plants [7-8]. Paddy fields are also a source of high levels of MeHg in downstream waterways [3, 9]. Therefore, understanding and predicting the accumulation of MeHg in paddy soils are critical for controlling the health risk of Hg. As a typical "hotspot" of Hg methylation, anaerobic paddy fields harbor the majority of the recognized Hg methylation microorganisms dominated by methanogens, sulfate-reducing bacteria and iron-reducing bacteria, which carry Hg methylating gene cluster *hgcAB* that is responsible for MeHg synthesis [10-14]. Recent studies have characterized Hg methylating communities and their associations with MeHg contents in paddy soils [15-16], which can be potentially used to predict MeHg production in the soil. However, other studies showed no significant relationship between Hg methylating genes expression and MeHg production in sediments [17-19], suggesting the limitations of predicting MeHg accumulation by Hg methylators alone. In addition, soil physical and chemical properties, such as pH, salinity, redox potential, organic carbon, sulfur, iron, and Hg availability, could also influence microbial Hg methylation. [20-24] Therefore, identifying new factors regulating Hg methylation is needed to strengthen our prediction of MeHg production in soil. It has been known that putative non-Hg methylators (defined as the taxa that do not carry *hgcAB*) could play important roles in affecting Hg bioavailability through physical and chemical processes [25-26], and also the activities of Hg methylators via microbial interactions [27]. Shifts of non-Hg methylating communities could also be one of factors controlling soil MeHg production, such as in boreal forest soils [28]. However, the effect of putative non-Hg methylating communities on MeHg production in paddy soils remains unclear.

It is well known that MeHg synthesis is subject to Hg availability and the activities of Hg methylating microorganisms [29], but factors affecting the two drivers could also be important for MeHg accumulation in paddy soils. For example, non-Hg methylators can adsorb Hg(II) on the cell surface or take Hg(II) inside the cell [30-31], resulting in a decrease of

free Hg(II) in the system. On the other hand, some non-Hg methylators (e.g. Hg resistant microbes) may reduce Hg(II) to Hg(0) that can evaporate from the soil to the atmosphere [26, 32]. These processes of Hg adsorption, uptake, and transformation can reduce the amount of bioavailable Hg for methylation in soils. Furthermore, some putative non-Hg methylating communities may also degrade the produced MeHg [33-35], thus changes in the abundance of soil microbial phylotypes are expected to influence the net MeHg production. In addition, non-Hg methylating communities may impact the prevalence and activities of Hg methylators in soils. Soil microorganisms strongly co-occur through multiple interactions (e.g. symbioses or sharing similar ecological function) within the microbial taxa [36-37], though the underlying mechanisms still remains unclear. Paddy soil is a typical anaerobic environment in which strong redox processes and syntrophic interactions among diverse microorganisms may both affect Hg methylation. For example, a recent incubation study suggested both positive and negative effects of non-Hg methylators on MeHg production by microbial syntrophy [27], but these effects were depend on the identity of microbial species. It thus implies that changes in soil microbial communities could influence Hg methylation in terms of Hg bioavailability, MeHg degradation and prevalence of Hg methylators. However, empirical evidence for such assumption is currently lacking.

Here, we aimed to identify the role of bacterial community composition in predicting MeHg production in paddy soils. In addition to Hg methylators, we hypothesized that some of putative non-Hg methylators may together govern the Hg methylation in paddy soils. To test our hypothesis, we conducted a national-scale study of MeHg production and the associated bacterial community in paddy soils across main rice-producing areas in China, with a wide gradient of Hg contents and soil properties. This relatively large sample distribution provided a unique opportunity to decipher the associations of both Hg and putative non-Hg methylators with MeHg production in the fields. We used MiSeq Illumina sequencing of 16S rRNA gene amplicons to characterize soil bacterial community composition, which were subsequently linked to MeHg production (reflected as MeHg content and %MeHg) [38-39]. We also ascertained the associations and co-occurrence patterns of key microbial taxa with Hg methylation using a combination of structure equation modelling, random forest analysis and network analysis. The identified link between specific soil

microbial phylotypes and MeHg contents from this study holds great potential in the in prediction of MeHg accumulation in paddy fields.

Materials and methods

Site description and sample collection

Soil samples were collected from the major rice-producing areas across twelve provinces from northeast to southwest in China (Figure S1), which contribute more than 82% of total rice supply in the country. These areas have a wide range of climatic variables and represent most types of paddy soils in China. The mean annual precipitation and temperature ranged from 423 to 1861 mm and 2.9 ºC to 26.3ºC, respectively. In each rice-producing province, 3-6 representative samples were collected from paddy fields, while more intensive sampling was carried out in southwest China (i.e. Hunan and Guizhou provinces) because of serious Hg pollution in soils [4, 40]. Soil sampling were carried out immediately after rice harvest in 2017, when the soil was mostly saturated (>100 water holding capacity). At each rice paddy site, multiple soil samples (1 to 15 cm depth) were collected and then mixed as one sample, and a total of 141 soils samples were obtained. All soil samples were sealed in sterile plastic bag, and immediately shipped back to our laboratory on ice. Each soil sample was separated into two sub-samples. One sub-sample was immediately freeze-dried by a vacuum freeze drying system (FDU-1100, Tokyo Rikakikaico Ltd.) and used for total Hg (THg) and MeHg determination, as well as for microbial DNA extraction and bacterial community analyses, while another sub-sample was air-dried for analyses of soil properties. Basic soil properties are showed in Table S1 in the Supporting Information.

Analysis of Hg and geochemical characteristics

For Hg analysis, 0.20 g of each soil was digested with 10 mL mixed solution (2 mol L^{-1} HNO_3 and 4 mol L^{-1} HCl) in a Teflon tube at 100°C for 2 hours. The total amount of Hg in these extracts was determined via cold vapor atomic fluorescence spectrometry (CVAFS). The method detection limit was 0.2 ng L^{-1}. For MeHg analysis, 0.10 g of each soil was extracted using the $CuSO_4$–methanol solvent [41-42]. The amount of MeHg was determined using an automatic MeHg analyzer (CVAFS, Brooks Rand model III, Brooks Rand Laboratories, USA) according to US EPA Method 1630 [43]. Standard reference materials

including GSS-9 (THg: 32 ± 3 ng g^{-1}) and ERMCC580 (75.5 ± 3.7 ng g^{-1}) were used as control for THg and MeHg quantification, respectively. Typical recoveries of THg and MeHg from the two reference materials were between 90.7 − 95.2% and 86.2−107.6%, respectively. Analytical blanks were also included in the analysis for quality control. Soil pH was determined on a fresh soil to water ratio of 1: 2.5 using a Delta pH-meter, and soil organic carbon (SOC) was measured using the K_2CrO_7 oxidation titration method [44]. Total carbon (TC) and total nitrogen (TN) in soils were determined on a LECO TureMac Macro CN analyzer (LECO, St. Joseph, MI, USA).

Soil DNA extraction and microbial community analysis

Soil DNA was extracted using the MoBio PowerSoil DNA Isolation Kit (QIAGEN Inc., USA) following the manufacturer's instructions. Bacterial community composition was characterized by Illumina MiSeq sequencing of the V4 region of 16S rRNA gene amplified using the primer pairs of 338F/806R [36]. The 25 μl PCR reaction mixture consisted of 12.5 μl PremixTaq™ (Takara Biotechnology, Dalian, China), 1 μl of each primer (10 μM), and 2 μl of template DNA. The barcoded PCR products were purified using the Wizard® SV Gel and PCR Clean-Up System (Promega, San Luis Obispo, USA). The purified amplicons were equimolarly mixed, and 2 × 300 bp paired-end sequencing was carried out on an Illumina Miseq sequencer (Illumina Inc., San Diego, USA). Raw and paired-end sequences were quality filtered, assembled using the Fast Length Adjustment of Short Reads (FLASH). A chimera filtering approach UPARSE was employed as the Operational Taxonomic Unit (OTU or phylotype) picking strategy at 97% sequence similarity. Randomly subsampled OTU tables (20,000 reads per sample) were generated to correct possible biases introduced by unequal sequencing efforts. Representative sequences from individual OTUs generated in UPARSE were processed using the Quantitative Insights into Microbial Ecology (QIIME) pipeline. Shifts in the bacterial community composition were measured by using two axes of a non-metric multidimensional scaling (NMDS) analysis of Bray-Curtis dissimilarities. The taxonomic identity of all phylotypes was determined using the SILVA ribosomal RNA gene database project [45]. In addition, the abundances of Hg methylating genes (i.e. Delta-proteobacterial and archaeal *hgcA*) were quantified on an iCycler iQ5 thermocycler

(Bio-Rad, USA) using the clade-specific degenerate primer pairs ORNL-Delta-HgcA and ORNL-Archaea-HgcA, respectively [46]. More detailed information on the *hgcA* gene quantification is provided in the Supporting Information.

Microbial network analysis

We established co-occurrence networks to identify the underlying interactions/associations among bacterial taxa related to MeHg production in paddy soils across China. We only kept the taxa (bacterial families) which occurred in > 50% of all the samples. We then calculated all pairwise Spearman's rank correlations (ρ) between all soil bacterial taxa using the "psych" package of the R statistical software (http://cran.r-project.org/) and focused exclusively on positive correlations as they provide information on bacterial taxa that may respond similarly to environmental conditions. We considered a co-occurrence to be robust only if the Spearman's correlation coefficient was > 0.50 and $P < 0.01$. Further, only taxa that were significantly correlated with potential Hg methylators were kept, making it relatively easy to decipher the associations between Hg and non-Hg methylators in the paddy soils. The network was visualized with the interactive platform Gephi [47]. Finally, we used default parameters from the interactive platform Gephi to identify ecological clusters (i.e., modules) of the selected soil taxa strongly interacting with each other [36].

Statistical analysis

We used structural equation modelling (SEM) to evaluate effects of bacterial community composition (axes of NMDS ordinations), the abundance of Delta-proteobacterial and archaeal *hgcA* gene on soil MeHg content and the potential methylation efficiency (as %MeHg), while the effects of soil properties (pH, SOC, C: N) were also considered. The use of SEM is particularly useful in investigating the complex relationships among predictors commonly found in natural ecosystems, as it allows the partitioning of causal influences among multiple variables and the separation of direct and indirect effects of model predictors [48]. The aim of SEM used in our study was to identify associations of soil microbial attributes to Hg methylation after accounting for other abiotic factors. We first established a *priori* model according to our current knowledge of the impact of environmental variables on soil

Hg methylation (MeHg content and %MeHg) (Figure S2). The data matrix was fitted to the model using the maximum-likelihood estimation method. Since there is no universally accepted single test of overall goodness of fit for SEM, we thus used the Chi-square test (χ^2; the model has a good fit when $0 \leq \chi^2/\text{d.o.f} \leq 2$ and $0.05 < p \leq 1.00$) and the root mean square error of approximation (RMSEA; the model has a good fit when RMSEA $0 \leq \text{RMSEA} \leq 0.05$ and $0.10 < p \leq 1.00$ [49]. The SEM analyses were performed using AMOS 21.0 (SPSS Inc., Chicago, IL, USA). The standardized total effects of the included variables on MeHg content and %MeHg were subsequently calculated to aid the interpretation of the SEMs. We also used Spearman's correlations to evaluate the correlations between MeHg production and all the examined soil microbial and geochemical attributes.

We then conducted a classification random forest analysis to identify significant ($P < 0.05$) microbial predictors (bacterial families and genera) of MeHg production. The goal of these analyses was to test which taxa are significant predictors of MeHg content. To estimate the importance of these microbial taxa, we compared the percentage increases in the MSE (mean squared error) of MeHg production: higher MSE% values imply more important variables [50]. These analyses were conducted using the "rfPermute" package of the R statistical software (http://cran.r-project.org/). After that, we used linear regressions to evaluate the direction of relationships between the relative abundance of identified taxa and MeHg concentrations. We then applied the two most common regression models [51] (linear and quadratic) to evaluate the relationships between the relative abundance of major identified taxa and soil MeHg contents.

RESULTS

Net MeHg production in paddy soils across China

Our results showed relatively high THg and MeHg contents in paddy soils from Guizhou and Hunan Provinces, ranging from 0.02 to 52.37 μg g^{-1} and 0.11 to 8.36 ng g^{-1}, respectively (Figure 1, Figure S3). However, we found high ratio of MeHg to THg (%MeHg) in the soils in Guangdong province and northeast China (Liaoning and Heilongjiang provinces) where the total Hg contents were relatively low. On the contrary, the highly Hg-impacted areas such as Guizhou and Hunan provinces in southwest China had relatively

low %MeHg in the soil, but with higher MeHg contents. Accordingly, we found a positive relationship between total Hg and MeHg in paddy soils across China, while the total Hg was negatively correlated with soil %MeHg (Figure S4a, b). In addition, MeHg content was positively correlated with spatial variation (longitude and latitude), soil pH, but negatively correlated with soil C: N and SOC ($p < 0.05$, Table S2).

Microbial community as indicators of MeHg production in paddy soils

Our SEMs suggested bacterial community composition has strong associations with MeHg contents and %MeHg even considering the edaphic factors (Figure 2a, b). Similar results were observed when we analyzed the total standardized effects (i.e. sum of directs and indirect effects) of the significant factors on soil %MeHg (Figure S5). Linear regression analysis further showed strong correlations ($p < 0.001$) of the bacterial community composition (two axes of NMDS ordinations) with MeHg contents (Figure 2c, d). No significant effect ($p > 0.05$) of Delta-proteobacterial *hgcA* abundance on MeHg was found, whereas the abundance of archaeal *hgcA* had a negative correlation ($p < 0.05$) with MeHg contents. Importantly, our SEM indicated that the effects of soil properties, such as SOC, pH, C: N on MeHg or %MeHg were mainly driven indirectly by changes in bacterial community composition and archaeal *hgcA* abundance.

Bacterial taxa as predictors of MeHg production in paddy soils

We used random forest modeling to identify statistically significant ($p < 0.05$) bacterial taxa of soil MeHg production (Figure 3a), and the results showed that 103 families were significantly ($p < 0.05$) associated with MeHg content (Table S3). Most of these predictive families are diverse non-Hg methylators (only two potential Hg methylators), with the *Xanthomonadales_Incertae_Sedis* as the strongest predictor. These identified bacterial families predominately affiliated within phyla/classes *Gammaproteobacteria*, *Deltaproteobacteria*, *Betaproteobacteria*, *Alphaproteobacteria*, *Actinobacteria*, *Bacteroidetes,* and *Firmicutes*. Some of these predictive families had significant relationships with soil MeHg contents (Figure 3b). For example, the relative abundances of unclassified *Xanthomonadale*, *Chitinophagaceae Bacteroidetes*_BD2-2_fa had strong positive correlations with the MeHg content, while the relative abundance of

*Actinobacteria*_PeM15_fa has a quadratic relationship with MeHg. In addition, a positive relationship between the relative abundance of the Hg methylating group *Ruminococcaceae* and MeHg content was observed in the soil (Figure S6).

Co-occurrence patterns of putative non-Hg and Hg methylators

The links between Hg and putative non-Hg methylators were visualized as association networks based on strong correlations (correlation coefficient > 0.5, $p < 0.01$). A total of 94 putative non-Hg methylators and 13 potential Hg methylators families strongly co-occurred in the paddy soils (Figure 4a, Table S4). Particularly, the Hg methylating group *Ruminococcaceae* showed the strongest correlation with multiple putative non-Hg methylators, whereas the remaining Hg methylators were correlated with Hg methylators and the putative non-Hg methylators. Strikingly, we found some potential syntrophic interactions between Hg methylators such as *Syntrophobacteraceae*, *Syntrophaceae*, *Syntrophorhabdaceae* and putative non-Hg methylators. The network was clearly separated into six clusters (Figure 4b, Table S4), and most of the modules contained both Hg and non-Hg methylators. Among them, module 1 contained the most diverse methylators, while module 3 was only comprised of the putative non-Hg methylators (Table S4).

DISCUSSION

Our study provides the first national-scale survey of relationships between Hg methylation and microbial community composition in paddy soils across China, which helps to improve our understanding of MeHg production and subsequent prediction of its potential health risk through the food webs. Our results suggest that variations of MeHg production are strongly linked to soil bacterial community composition even compared to abiotic factors such as soil pH and SOC. More importantly, our results suggest that putative non-Hg methylators could play a more important role than Hg methylators in predicting the changes of soil MeHg production. The combined use of MeHg content and %MeHg could also better reflect Hg methylation than any other single index, since the proxies including the ratio of methylation to demethylation (K_m/K_d), %MeHg and MeHg content all have limitations in accurately describing Hg methylation in situ [29].

It is surprising to find lower average MeHg content but higher %MeHg in northeast

China than that in southwest China where total Hg levels are higher, and this could be attributed to the low bioavailability of Hg in the soil. Numerous studies have suggested that inactive HgS is the dominant Hg species in soils of southwest China such as Guizhou and Hunan provinces, which was mainly derived from Hg-containing minerals (such as cinnabar) [40, 52]. Differently, anthropogenic emissions are the main sources of Hg in soils of northern regions, where the "new Hg" could have higher bioavailability compared to the "old Hg" [53]. Our results thus have important implications for future prediction of MeHg content in these rice-producing areas, considering the increasing Hg levels, especially in north China where industrial wastewater is the main source of Hg in the environment according to a recent study [54]. On the other hand, differences in distribution and composition of soil microbial community could also impact Hg bioavailability and the activities of Hg methylators, resulting in soil MeHg production to various extents across different regions. Therefore, monitoring of the dynamics of both Hg and non-Hg methylating communities could be particularly useful to improve prediction of MeHg production in paddy soils.

Our SEMs provide a systematic understanding of MeHg production, which allows the effect of microbial communities on soil Hg methylation to be examined separately from other environmental variables. As expected, our SEMs demonstrate strong associations of bacterial community composition with the MeHg content and %MeHg in the soil [28], implying shifts in soil bacterial community composition could result in predictable variations in MeHg production. Our findings also offer a new perspective that net MeHg production in natural ecosystems can be predicted based on the changes in microbial communities involving Hg and non-Hg methylators. This idea is further supported by the strong indirect effects of soil properties such as pH and SOC on MeHg production via soil microbial attributes, rather than in a direct manner. Thus, understanding the roles of putative non-Hg methylating communities should help better predict net MeHg content in paddy soils. Interestingly, the abundance of bacterial *hgcA* gene in paddy soils had no significant effect on MeHg content or %MeHg, which is consistent with a recent study showing no significant correlation between the *hgcA* abundance and MeHg content [36]. This could be attributed to the fact that DNA-based quantification of *hgcA* cannot explicate activities of the Hg methylators in the soils, as previously suggested. [17-19] In addition, the net MeHg content was governed by both

284 microbial Hg methylation and MeHg degradation. Unexpectedly, the SEM showed a negative
285 correlation between the abundance of archaeal *hgcA* and MeHg content, which could be
286 related to potential demethylation of MeHg by the methanogens.[33] These results demonstrate
287 limitations of *hgcA* abundance in explaining the Hg methylation, and identifying the
288 associations between putative non-Hg methylators and Hg methylation can be a critical step
289 towards accurate prediction of MeHg accumulation in paddy soils. We claim that cautions
290 should be taken when using *hgcA* gene abundance to predict MeHg production, and specific
291 niche conditions and total microbial community should also be considered.

292 We found distinct soil bacterial community compositions across different rice-growing
293 regions in China (Figure S7), which could be one of main factors leading to the high
294 variations of MeHg among the soils according to our SEM analysis. Although many previous
295 studies have been devoted to examining the link between Hg methylating communities and
296 MeHg production in the environment [12, 16], much less attention was paid to the effect of
297 non-Hg methylating communities on MeHg production. In the present study, we identified
298 several key bacterial taxa (families) capable of predicting the soil MeHg variations,
299 suggesting their potential roles in regulating MeHg accumulation either through MeHg
300 formation or MeHg degradation. However, only one potential Hg methylator was among the
301 identified top 10 microbial predictors of soil MeHg content, suggesting a more important role
302 of putative non-Hg methylators in predicting MeHg production compared to Hg methylators.
303 For example, our random forest modeling suggested family
304 *Xanthomonadales*_Incertae_Sedis, *Chitinophagaceae* and *Bacteroidetes*_BD2.2_fa as the top
305 three predictors of MeHg, and this result still remained even when carrying out a second run
306 of random forest analysis based on further genus level (Table S5). Furthermore, we also
307 experimentally confirmed the inability of the predictive families for methylation using the
308 representatives of *Xanthomonadale* and (i.e. *Chitinophaga dinghuensis* and *Xanthomonas*
309 *sacchari*, respectively) (Figure S8), and this further supports our hypothesis that the putative
310 non-Hg methylators could impact MeHg production indirectly. However, we cannot rule out
311 the possibility that other putative non-methylators may methylate Hg via other pathways [55].

312 The strong relationships between the putative non-Hg methylators and Hg methylation
313 also support our argument that these taxa could be used as potential bio-indicators of Hg

methylation in paddy soils. For example, the relative abundances of the main predictive taxa are much higher in southwest China than those in northeast China (Figure S9), which is in line with the observations of relatively high MeHg contents in southwest China. Our findings agree with a recent study suggested that *Xanthomonadales* could be tolerant to Hg, although the underlying mechanism remains unclear.[56] A previous study also identified a positive correlation between the abundance of *Chitinophagaceae* spp. and soil THg [36], implying its important role in providing Hg for methylation. However, no more evidence can be found regarding the associations between the putative non-Hg methylators and Hg methylation. Non-Hg methylators are dominant microbial communities in paddy soils, and they can potentially influence the Hg bioavailability and the prevalence of Hg methylators through microbial interactions. Further evidence can be collected by investigating the effects of those taxa on microbial Hg methylation assays under lab-controlled conditions. In addition, it is not surprising to find the significant positive correlation between methylators *Ruminococcaceae* abundance and MeHg content (Figure S6), and this is in line with the traditional argument that Hg methylators are important predictors of MeHg production in the paddy soils [6].

Interestingly, we find strong co-occurrence between taxa predictors including both putative non-Hg and Hg methylators, which can partially explain the role of the non-Hg taxa in Hg methylation. For example, top microbial predictors such as *Chitinophagaceae*, *Bacteroidetes*_BD2.2_fa and *Actinobacteria*_PeM15_fa strongly co-occurred with the potential methylators *Ruminococcaceae*, *Spirochaetaceae* and *Ruminococcaceae*, respectively. The significant co-occurrence between Hg and putative non-Hg methylators is not a surprise, given the fact that bacterial taxa could exhibit predictable co-occurrence patterns in soil [37]. Assembly patterns of soil microorganisms may illustrate they could be symbiotic or syntrophic relationship, or they share some common functions (Figure 4a), highlighting the importance of non-Hg methylators for Hg methylation [27]. For example, the potential methylator *Ruminococcaceae* and *Desulfarculaceae* co-occurred with multi-taxa including putative non-Hg methylators, though the underlying co-occurrence mechanisms remain uncovered. These results are also consistent with the general idea that soil microorganisms tended to co-occur more than expected [57].

In all likelihood, the co-occurrence patterns of Hg methylators with putative non-Hg

methylators can be useful for improving MeHg prediction in paddy soil, especially under the circumstance when no significant relationship was found between Hg methylating communities and the MeHg content. However, some of the co-occurrence patterns could reveal interesting ecological patterns for the microbial taxa that have not been well studied, which may advance our understanding of associations between putative non-Hg and Hg methylators. Our results also demonstrate the insufficiency of only using Hg methylators to predict MeHg in paddy soil, and effects of putative non-Hg methylators on Hg methylation should be considered.

In conclusion, this study provides compelling evidence that variations of MeHg production in paddy soils at a national spatial scale are strongly associated with changes in soil microbial community composition. We identified multiple microbial taxa capable of predicting MeHg variations in paddy soils across China, and highlight the important role of putative non-Hg methylators in soil Hg methylation. These results are significant as they reveal, in unprecedented detail, co-occurrence between Hg methylators and putative non-Hg methylators in paddy soils. These findings advance our understanding of Hg methylation in paddy systems, but deep mechanisms still need to be uncovered in future studies. This work may also provide new insights in managing MeHg pollution in response to increasing Hg levels from anthropogenic activities and global environment changes [42, 58-59].

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ASSOCIATED CONTENT

Supporting Information and Data Availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Materials. This material is available free of charge via the Internet at http://pubs.acs.org.

REFERENCES

1. Driscoll, C. T.; Mason, R. P.; Chan, H. M.; Jacob, D. J.; Pirrone, N., Mercury as a global pollutant: sources, pathways, and effects. *Environl. Sci. Technol.* **2013,** *47*, (10), 4967-4983.
2. Jonsson, S.; Andersson, A.; Nilsson, M. B.; Skyllberg, U.; Lundberg, E.; Schaefer, J. K.; Åkerblom, S.; Björn, E., Terrestrial discharges mediate trophic shifts and enhance methylmercury accumulation in estuarine biota. *Sci. Adv.* **2017,** *3*, (1), e1601239.
3. Hsu-Kim, H.; Eckley, C. S.; Achá, D.; Feng, X.; Gilmour, C. C.; Jonsson, S.; Mitchell, C. P. J., Challenges and opportunities for managing aquatic mercury pollution in altered landscapes. *Ambio* **2018,** *47*, (2), 141-169.
4. Feng, X.; Li, P.; Qiu, G.; Wang, S.; Li, G.; Shang, L.; Meng, B.; Jiang, H.; Bai, W.; Li, Z.; Fu, X., Human exposure to methylmercury through rice intake in mercury mining areas, Guizhou province, China. *Environl. Sci. Technol.* **2008,** *42*, (1), 326-332.
5. Rothenberg, S. E.; Anders, M.; Ajami, N. J.; Petrosino, J. F.; Balogh, E., Water management impacts rice methylmercury and the soil microbiome. *Sci. Total Environ.* **2016,** *572*, 608-617.
6. Liu, Y.-R.; Yu, R.-Q.; Zheng, Y.-M.; He, J.-Z., Analysis of the microbial community structure by monitoring an Hg methylation gene (hgcA) in paddy soils along an Hg Gradient. *Appl. Environ. Microbiol.* **2014,** *80*, (9), 2874-2879.
7. Meng, B.; Feng, X.; Qiu, G.; Liang, P.; Li, P.; Chen, C.; Shang, L., The process of methylmercury accumulation in rice (Oryza sativa L.). *Environl. Sci. Technol.* **2011,** *45*, (7), 2711-2717.
8. Li, Y.; Zhao, J.; Zhang, B.; Liu, Y.; Xu, X.; Li, Y.-F.; Li, B.; Gao, Y.; Chai, Z., The influence of iron plaque on the absorption, translocation and transformation of mercury in rice (Oryza sativa L.) seedlings exposed to different mercury species. *Plant Soil.* **2016,** *398*, (1), 87-97.
9. Rothenberg, S. E.; Windham-Myers, L.; Creswell, J. E., Rice methylmercury exposure and mitigation: a comprehensive review. *Environ. Res.* **2014,** *133*, 407-423.
10. Parks, J. M.; Johs, A.; Podar, M.; Bridou, R.; Hurt, R. A.; Smith, S. D.; Tomanicek, S. J.; Qian, Y.; Brown, S. D.; Brandt, C. C.; Palumbo, A. V.; Smith, J. C.; Wall, J. D.; Elias, D. A.; Liang, L., The genetic basis for bacterial mercury methylation. Science 2013, 339, (6125), 1332-1335.
11. Gilmour, C. C.; Podar, M.; Bullock, A. L.; Graham, A. M.; Brown, S. D.; Somenahally, A. C.; Johs, A.; Hurt, R. A.; Bailey, K. L.; Elias, D. A., Mercury methylation by novel microorganisms from new environments. *Environl. Sci. Technol.* **2013**, 47, (20), 11810-11820.
12. Bravo, A. G.; Zopfi, J.; Buck, M.; Xu, J.; Bertilsson, S.; Schaefer, J. K.; Poté, J.; Cosio, C., Geobacteraceae are important members of mercury-methylating microbial communities of sediments impacted by waste water releases. *ISME J.* **2018**, 12, 802-812.
13. Gionfriddo, C. M.; Tate, M. T.; Wick, R. R.; Schultz, M. B.; Zemla, A.; Thelen, M. P.; Schofield, R.; Krabbenhoft, D. P.; Holt, K. E.; Moreau, J. W., Microbial mercury methylation in Antarctic sea ice. *Nat. Microbiol.* **2016**, 1, 16127.
14. Podar, M.; Gilmour, C. C.; Brandt, C. C.; Soren, A.; Brown, S. D.; Crable, B. R.; Palumbo, A. V.; Somenahally, A. C.; Elias, D. A., Global prevalence and distribution of genes and microorganisms involved in mercury methylation. *Sci. Adv.* **2015**, 1, (9). e1500675.
15. Liu, Y.-R.; Johs, A.; Bi, L.; Lu, X.; Hu, H.-W.; Sun, D.; He, J.-Z.; Gu, B., Unraveling microbial communities associated with methylmercury production in paddy soils. *Environl. Sci. Technol.* **2018**, 52, (22), 13110-13118.
16. Ma, M.; Du, H.; Sun, T.; An, S.; Yang, G.; Wang, D., Characteristics of archaea and bacteria in rice rhizosphere along a mercury gradient. *Sci. Total Environ.* **2019**, 650, 1640-1651.

17. Bouchet, S.; Goñi-Urriza, M.; Monperrus, M.; Guyoneaud, R.; Fernandez, P.; Heredia, C.; Tessier, E.; Gassie, C.; Point, D.; Guédron, S.; Acha, D.; Amouroux, D., Linking microbial activities and low molecular weight thiols to Hg methylation in biofilms and periphyton from high altitude tropical lakes (Bolivian altiplano). *Environl. Sci. Technol.* **2018**, 52, 9758-9767.
18. Bravo, A. G.; Loizeau, J.-L.; Dranguet, P.; Makri, S.; Björn, E.; Ungureanu, V. G.; Slaveykova, V. I.; Cosio, C., Persistent Hg contamination and occurrence of Hg-methylating transcript (hgcA) downstream of a chlor-alkali plant in the Olt River (Romania). *Environ. Sci. Pollut. Res.* **2016**, 23, (11), 10529-10541.
19. Paranjape, A. R.; Hall, B. D., Recent advances in the study of mercury methylation in aquatic systems. *FACETS* **2017**, 2, (1), 85-119.
20. Rothenberg, S. E.; Feng, X., Mercury cycling in a flooded rice paddy. Journal of Geophysical Research: *Biogeosciences* **2012**, 117, G03003, 10.1029/2011JG001800.
21. Zhao, L.; Qiu, G.; Anderson, C. W. N.; Meng, B.; Wang, D.; Shang, L.; Yan, H.; Feng, X., Mercury methylation in rice paddies and its possible controlling factors in the Hg mining area, Guizhou province, Southwest China. *Environ. Pollut.* **2016**, *215,* 1-9.
22. Ullrich, S. M.; Tanton, T. W.; Abdrashitova, S. A., Mercury in the aquatic environment: a review of factors affecting methylation. *Cri.t Rev. Env. Sci. Tech.* **2001**, 31, 241-293.
23. Benoit, J. M.; Gilmour, C. C.; Mason, R. P., The influence of sulfide on solid-phase mercury bioavailability for methylation by pure cultures of Desulfobulbus propionicus (1pr3). *Environl. Sci. Technol.* **2001**, 35, (1), 127-132.
24. Sun, T.; Ma, M.; Du, H.; Wang, X.; Zhang, Y.; Wang, Y.; Wang, D., Effect of different rotation systems on mercury methylation in paddy fields. *Ecotoxicol. Environ. Saf.* **2019**, 182, 109403.
25. Barkay, T.; Wagner-Dobler, I., Microbial transformations of mercury: potentials, challenges, and achievements in controlling mercury toxicity in the environment. *Adv. Appl. Microbiol.* **2005,** *57*, (1), 1-52.
26. Grégoire, D. S.; Poulain, A. J., Shining light on recent advances in microbial mercury cycling. *FACETS* **2018,** 3, (1), 858-879.
27. Yu, R.-Q.; Reinfelder, J. R.; Hines, M. E.; Barkay, T., Syntrophic pathways for microbial mercury methylation. *ISME J.* **2018,** *12*, (7), 1826-1835.
28. Xu, J.; Buck, M.; Eklöf, K.; Ahmed, O. O.; Schaefer, J. K.; Bishop, K.; Skyllberg, U.; Björn, E.; Bertilsson, S.; Bravo, A. G., Mercury methylating microbial communities of boreal forest soils. *Sci. Rep.* **2019,** *9*, (1), 518.
29. Regnell, O.; Watras, C. J., Microbial mercury methylation in aquatic environments: a critical review of published field and laboratory studies. *Environ. Sci. Technol.* **2018**, 53, 4-19.
30. Szczuka, A.; Morel, F. M. M.; Schaefer, J. K., The effect of thiols, zinc, and redox conditions on Hg uptake in Shewanella oneidensis. *Environ. Sci. Technol.* **2015**, 49, 7432-7438.
31. Lin, H.; Hurt, R. A.; Johs, A.; Parks, J. M.; Morrell-Falvey, J. L.; Liang, L.; Elias, D. A.; Gu, B., Unexpected effects of gene deletion on interactions of mercury with the methylation-deficient mutant ΔhgcAB. *Environ. Sci. Technol. Lett.* **2014,** *1*, (5), 271-276.
32. Frossard, A.; Hartmann, M.; Frey, B., Tolerance of the forest soil microbiome to increasing mercury concentrations. *Soil Biol. Biochem.* **2017,** *105*, 162-176.
33. Kronberg, R.-M.; Schaefer, J. K.; Björn, E.; Skyllberg, U., Mechanisms of methyl mercury net degradation in alder swamps: the role of methanogens and abiotic processes. *Environ. Sci. Technol. Lett.* **2018**, 5: 220-225.
34. Lu, X.; Gu, W.; Zhao, L.; Haque, M. F. U.; DiSpirito, A. A.; Semrau, J. D.; Gu, B., Methylmercury uptake

and degradation by methanotrophs. *Sci. Adv.* **2017,** *3*, (5), e1700041.

35. Schaefer, J. K.; Yagi, J.; Reinfelder, J. R.; Cardona, T.; Ellickson, K. M.; Tel-Or, S.; Barkay, T., Role of the Role of the bacterial organomercury lyase (MerB) in controlling methylmercury accumulation in mercury-contaminated natural waters. *Environ. Sci. Technol.* **2004,** *38*, (16), 4304-4311.
36. Liu, Y.-R.; Delgado-Baquerizo, M.; Bi, L.; Zhu, J.; He, J.-Z., Consistent responses of soil microbial taxonomic and functional attributes to mercury pollution across China. *Microbiome* **2018,** *6*, (1), 183.
37. Barberán, A.; Bates, S. T.; Casamayor, E. O.; Fierer, N., Using network analysis to explore co-occurrence patterns in soil microbial communities. *ISME J.* **2012,** *6*, (2), 343-351.
38. Skyllberg, U.; Drott, A.; Lambertsson, L.; Björn, E.; Karlsson, T.; Johnson, T.; Heinemo, S.-Å.; Holmström, H., Net methylmercury production as a basis for improved risk assessment of mercury-contaminated sediments.*Ambio* **2007,** *36*, (6), 437-442.
39. Drott, A.; Lambertsson, L.; Björn, E.; Skyllberg, U., Do potential methylation rates reflect accumulated methyl mercury in contaminated sediments? *Environ. Sci. Technol.* **2008,** *42*, (1), 153-158.
40. Meng, M.; Li, B.; Shao, J.-J.; Wang, T.; He, B.; Shi, J.-B.; Ye, Z.-H.; Jiang, G.-b., Accumulation of total mercury and methylmercury in rice plants collected from different mining areas in China. *Environ. Pollut.* **2014,** *184*, 179-186.
41. Kronberg, R. M.; Drott, A.; Jiskra, M.; Wiederhold, J. G.; Björn, E.; Skyllberg, U., Forest harvest contribution to Boreal freshwater methyl mercury load. *Global Biogeochem. Cy.* **2016**, 30,825-843.
42. Yang, Z.; Fang, W.; Lu, X.; Sheng, G.-P.; Graham, D. E.; Liang, L.; Wullschleger, S. D.; Gu, B., Warming increases methylmercury production in an Arctic soil. *Environ. Pollut.* **2016,** *214*, 504-509.
43. USEPA, US EPA (United States Environmental Protection Agency), 2001. Method 1630: Methyl mercury in water by distillation, aqueous ethylation, purge and trap, and CVAFS. **2011**.
44. Walkley, A.; Black, I. A., An examination of the Degtjareff method for determining soil organic matter, and a proposed modification of the chromic acid titration method. *Soil Sci.* **1934,** *37*, (1), 29-38.
45. Quast, C.; Pruesse, E.; Yilmaz, P.; Gerken, J.; Schweer, T.; Yarza, P.; Peplies, J.; Glöckner, F. O., The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Res.* **2013,** *41*, (D1), D590-D596.
46. Christensen, G. A.; Wymore, A. M.; King, A. J.; Podar, M.; Hurt, R. A.; Santillan, E. U.; Soren, A.; Brandt, C. C.; Brown, S. D.; Palumbo, A. V.; Wall, J. D.; Gilmour, C. C.; Elias, D. A., Development and validation of broad-range qualitative and clade-specific quantitative molecular probes for assessing mercury methylation in the environment. *Appl Environ Microbiol.* **2016***, 82*, 6068-6078.
47. Bastian, M.; Heymann, S.; Jacomy, M., *Gephi: An Open Source Software for Exploring and Manipulating Networks*. 2009.
48. Grace, J. B., *Structural equation modeling and natural systems*. Cambridge University Press: 2006.
49. Delgado-Baquerizo, M.; Maestre, F. T.; Reich, P. B.; Jeffries, T. C.; Gaitan, J. J.; Encinar, D.; Berdugo, M.; Campbell, C. D.; Singh, B. K., Microbial diversity drives multifunctionality in terrestrial ecosystems. *Nat. Commun.* **2016,** *7*, 10541.
50. Archer, E., rfPermute: Estimate Permutation p-Values for Random Forest Importance Metrics. *R package* **2016,** *version 1.5.2.*
51. Laliberté E, Zemunik G, Turner BL (2014) Environmental filtering explains variation inplant diversity along resource gradients. *Science* **2014,** *345*,1602–1605.
52. Zhu, H.; Zhong, H.; Evans, D.; Hintelmann, H., Effects of rice residue incorporation on the speciation, potential bioavailability and risk of mercury in a contaminated paddy soil. *J. Hazard. Mater.* **2015,** *293*, (0), 64-71.

505 53. Jonsson, S.; Skyllberg, U.; Nilsson, M. B.; Lundberg, E.; Andersson, A.; Björn, E., Differentiated
506 availability of geochemical mercury pools controls methylmercury levels in estuarine sediment and biota.
507 *Nat. Commun.* **2014,** *5*, 4624.
508 54. Liu, M.; Zhang, W.; Wang, X.; Chen, L.; Wang, H.; Luo, Y.; Zhang, H.; Shen, H.; Tong, Y.; Ou, L.; Xie,
509 H.; Ye, X.; Deng, C., Mercury release to aquatic environments from anthropogenic sources in China from
510 2001 to 2012. *Environ Sci Technol.* **2016**,*50*. 8169-8177.
511 55. Xiang, Y.; Wang, Y.; Shen, H.; Wang, D., The draft genome sequence of pseudomonas putida Strain
512 TGRB4, an aerobic bacterium capable of producing methylmercury. Curr Microbiol **2019**,
513 https://doi.org/10.1007/s00284-019-01670-357.
514 56. Frey, B.; Rieder, S. R., Response of forest soil bacterial communities to mercury chloride application. *Soil*
515 *Biol. Biochem.* **2013**, 65, 329-337.
516 57. Ma, B.; Zhao, K.; Lv, X.; Su, W.; Dai, Z.; Gilbert, J. A.; Brookes, P. C.; Faust, K.; Xu, J., Genetic
517 correlation network prediction of forest soil microbial functional organization. *ISME J.* **2018**, *12*, 2492.
518 58. Kwon, S.; Selin, N.; Giang, A.; Karplus, V.; Zhang, D., Present and future mercury concentrations in
519 Chinese rice: Insights from modeling. *Global Biogeoche Cy.* **2018**, *32*, 437-462.
520 59. Obrist, D.; Kirk, J. L.; Zhang, L.; Sunderland, E. M.; Jiskra, M.; Selin, N. E., A review of global
521 environmental mercury processes in response to human and natural perturbations: Changes of emissions,
522 climate, and land use. *Ambio* **2018,** *47*, (2), 116-140.

FIGURE LEGENDS:

Figure 1. Location and MeHg production (**a**, MeHg concentrations and **b**, % MeHg) in 141 paddy soils across China.

Figure 2. Associations of soil microbial community with % MeHg. **(a, b)** The structural equation modelling (SEM) showing the direct or indirect effects of soil properties and microbial attributes on MeHg concentration and %MeHg, respectively. The thickness of the arrow represents the strength of the relationship when significant, while no arrow is showed when the effect is not significant. Numbers adjacent to arrows are path coefficients with significant levels. R^2 denotes the proportion of variance explained, *$p < 0.05$ and ** $p < 0.01$; **(c)** and **(d)** linear correlations between bacterial community composition (axes of NMDS ordinations) and MeHg concentrations.

Figure 3. (a) Random forest analyses identifying the top 10 significant ($p < 0.05$) predictors (at family level) of soil MeHg concentrations; **(b)** relationships between the relative abundance of selected taxa and soil MeHg concentrations.

Figure 4. The networks showing the co-occurrence patterns among the putative non-Hg methylators and Hg methylators in the paddy soils. **(a)**The green nodes represent Hg methylators and the red ones represent putative Hg-non methylators. **(b)** The nodes coded with different colours represent different classes of modules. The edges connecting nodes indicate strong ($r > 0.5$) and significant ($p < 0.01$) correlations between nodes.

Fig. 1

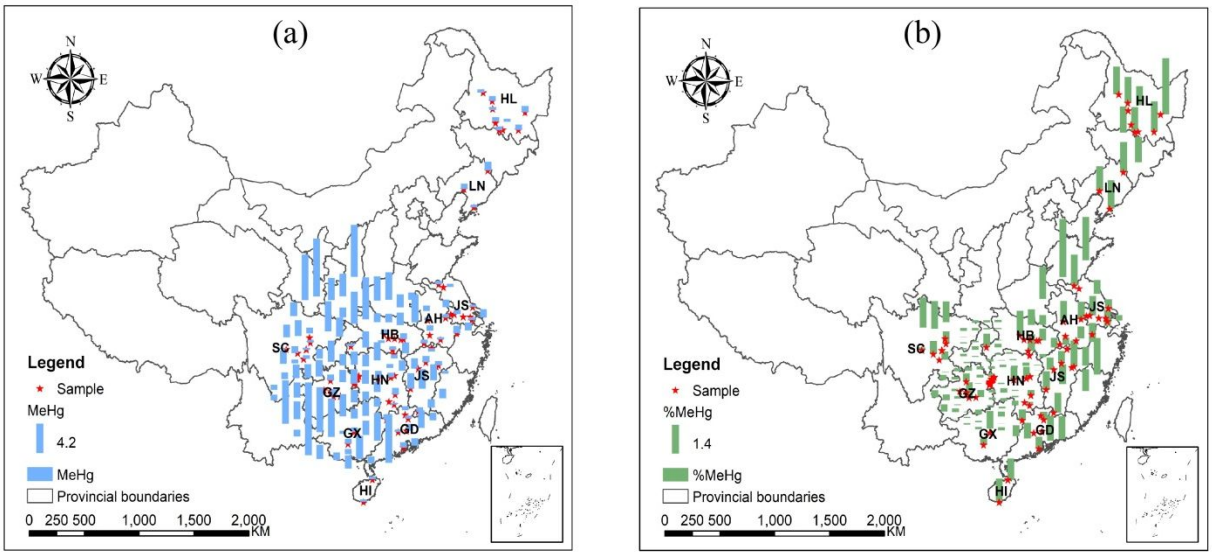

(a)
N
W
E
S
HL
LN
JS
AH
HB
SC
HN
JS
GZ
GX
GD
HI
Legend
Sample
MeHg
4.2
MeHg
Provincial boundaries
0 250 500 1,000 1,500 2,000
KM
(b)
N
W
E
S
HL
LN
JS
AH
HB
SC
HN
JS
GZ
GX
GD
HI
Legend
Sample
%MeHg
1.4
%MeHg
Provincial boundaries
0 250 500 1,000 1,500 2,000
KM

Fig. 2

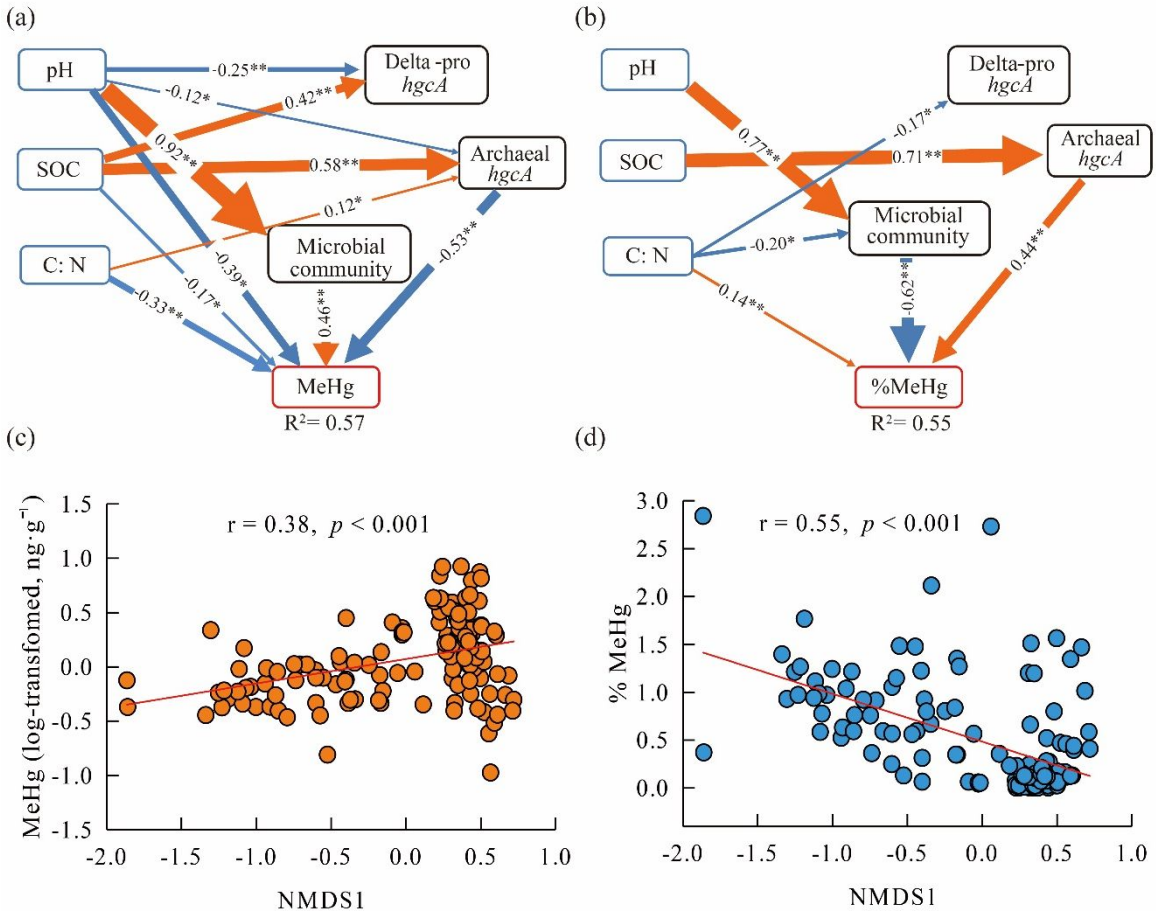

(a)
pH
SOC
C: N
Delta -pro hgcA
Archaeal hgcA
Microbial community
MeHg
R²= 0.57
-0.25**
-0.12*
0.42**
0.92**
0.58**
0.12*
-0.39*
-0.17*
-0.33**
0.46**
-0.53**
(b)
pH
SOC
C: N
Delta-pro hgcA
Archaeal hgcA
Microbial community
%MeHg
R²= 0.55
0.77**
-0.17*
0.71**
-0.20*
-0.62**
0.14**
0.44**
(c)
r = 0.38, p < 0.001
MeHg (log-transfomed, ng·g⁻¹)
NMDS1
(d)
r = 0.55, p < 0.001
% MeHg
NMDS1

Fig. 3

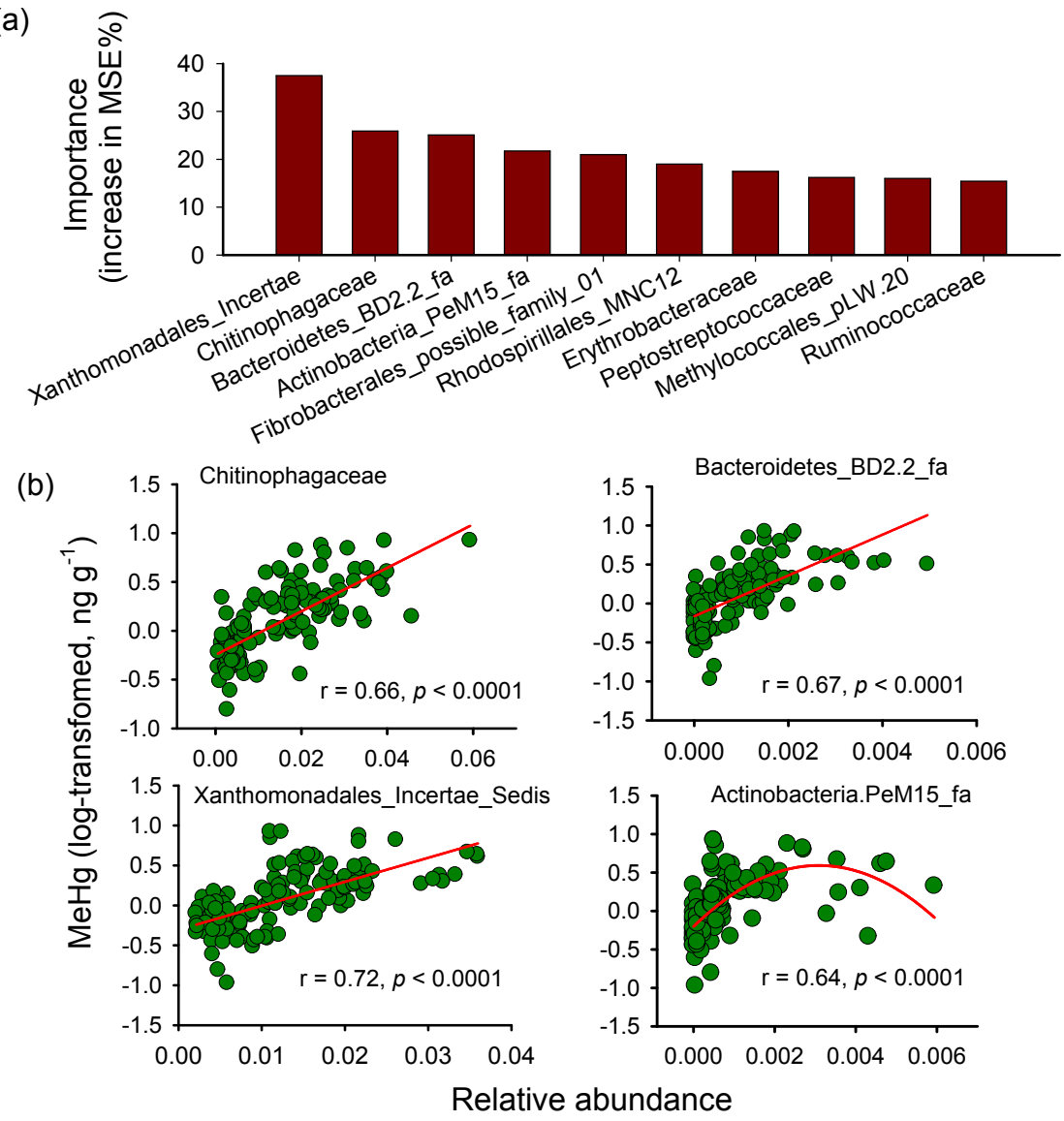
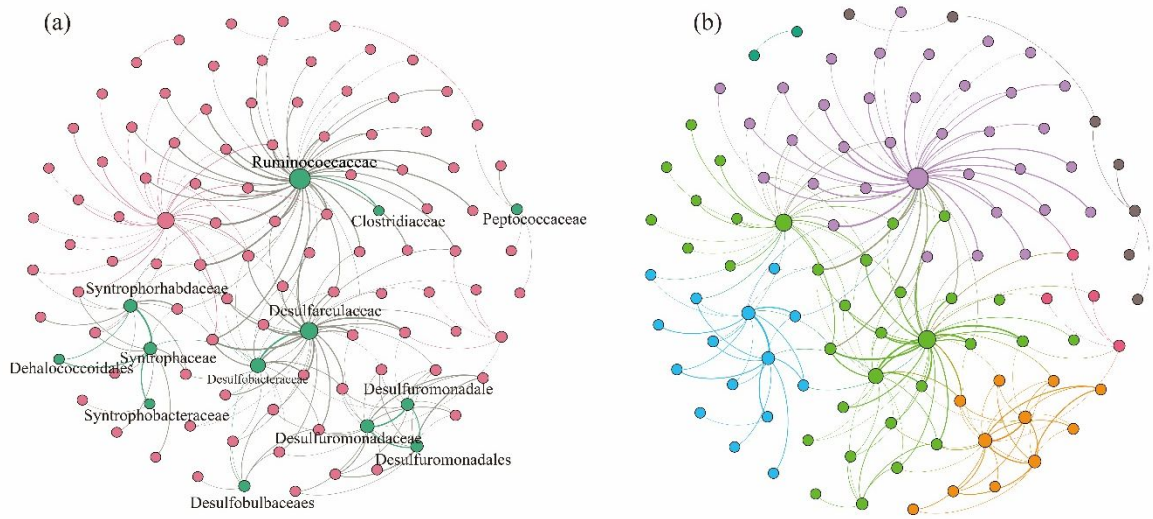
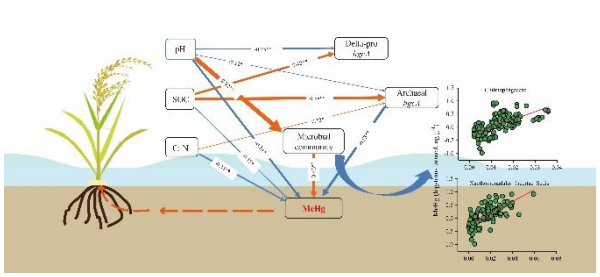

(a)
Importance (increase in MSE%)
40
30
20
10
0
Xanthomonadales_Incertae
Chitinophagaceae
Bacteroidetes_BD2.2_fa
Actinobacteria_PeM15_fa
Fibrobacterales_possible_family_01
Rhodospirillales_MNC12
Erythrobacteraceae
Peptostreptococcaceae
Methylococcales_pLW.20
Ruminococcaceae
(b)
MeHg (log-transfomed, ng g-1)
Chitinophagaceae
r = 0.66, p < 0.0001
Bacteroidetes_BD2.2_fa
r = 0.67, p < 0.0001
Xanthomonadales_Incertae_Sedis
r = 0.72, p < 0.0001
Actinobacteria.PeM15_fa
r = 0.64, p < 0.0001
Relative abundance

Fig. 4


(a)
Ruminococcaceae
Clostridiaceae
Peptococcaceae
Syntrophorhabdaceae
Desulfarculaceae
Syntrophaceae
Dehalococcoidales
Desulfobacteraceae
Desulfuromonadale
Syntrophobacteraceae
Desulfuromonadaceae
Desulfuromonadales
Desulfobulbaceaes
(b)



TOC Art